

Divyansh Shukla

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SUMMARY

Computer Science undergraduate (AI & ML) who builds deep learning systems from first principles in PyTorch — diffusion transformers, 2D rotary attention, and rectified flow written from scratch — plus object detection and reinforcement learning pipelines trained on cloud GPUs against benchmarked metrics. Seeking machine learning and research engineering roles.

EDUCATION

Vellore Institute of Technology

B.Tech, Computer Science and Engineering — Artificial Intelligence & Machine Learning

CGPA: 9.03 / 10

TECHNICAL SKILLS

Languages: Python, C++, CUDA

ML Frameworks: PyTorch, PyTorch Lightning, Hugging Face Transformers, TRL, Ultralytics (YOLO), Stable-Baselines3, vLLM

Deep Learning: diffusion models, flow matching, transformers, attention mechanisms (RoPE, AdaLN-Zero), classifier-free guidance, object detection, reinforcement learning (PPO), vision-language models

Simulation & Robotics: MuJoCo, Gymnasium, OpenCV, Raspberry Pi, embedded/edge inference

Infrastructure: Modal (cloud GPU training), Weights & Biases, Hydra, FastAPI, Git, Linux, NumPy, pandas, HDF5

PROJECTS

Thermal Object Detection for ADAS

github.com/divyanshuklai/car-safety-software-ADAS

- Built an end-to-end detection pipeline on the FLIR ADAS v2 thermal dataset, training YOLOv8n over 6 road-safety classes with resumable multi-hour GPU runs orchestrated on Modal using persistent volumes and W&B logging.
- Wrote a COCO-to-YOLO converter that symlinks rather than copies images to avoid duplicating the dataset, fuzzy-matching FLIR category aliases onto contiguous class IDs.
- Tuned the recipe for single-channel thermal input by disabling hue/saturation augmentation and raising classification loss weight against background confusion; evaluation reports per-class AP50 and AP50-95 against a fixed baseline (mAP50 0.59, mAP50-95 0.36).

RoPE-DiT: Diffusion Transformer with 2D RoPE

github.com/divyanshuklai/Custom2DRoPEDiT

- Implemented a ~5M-parameter Diffusion Transformer from scratch: patchifier, timestep and class-label embedders, AdaLN-Zero conditioning, and a custom multi-head attention module.
- Designed a true 2D rotary positional embedding that splits the attention head dimension into row and column halves and rotates queries/keys by each patch's (height, width) coordinates, encoding relative 2D distance rather than flattened 1D or absolute positions.
- Trained on CIFAR-10 with a rectified flow objective (MSE against the velocity field $v = x_1 - x_0$), sampling via a hand-written Euler ODE solver with classifier-free guidance.

Zero-Shot ICE RMA: Quadruped Locomotion RL

github.com/divyanshuklai/zero-shot-ice-rma

- Built a MuJoCo environment for the Unitree Go1 quadruped with a 30-dim proprioceptive observation space and 12-dim joint-position action space, plus fractal terrain for domain randomization.
- Implemented the bioenergetics reward from the Rapid Motor Adaptation paper (forward velocity, lateral/rotation penalty, work, impact, smoothness) with curriculum scheduling that anneals penalty weights early in training.
- Diagnosed a failed 42M-step PPO baseline, tracing exploding value loss and absent gait discovery to unnormalized observations, penalty dominance, and missing previous-action context.

Fashion Search: Conversational Product Discovery

github.com/divyanshuklai/fashion_search

- Built a FastAPI chat app turning natural-language fashion queries into product cards, chaining Tavily web search, Gemini 2.5 Flash structured extraction, and BeautifulSoup scraping across 10 Indian e-commerce stores.
- Engineered a priority-based image and price extractor (JSON-LD schema → Open Graph meta tags → keyword-matched image tags → fallback), documenting baseline failure modes to benchmark an in-progress agentic variant.

Smart Waste Segregator

divyanshuklai.github.io/projects

- Built an autonomous bin classifying waste into four categories (organic, metal, plastic, glass) fully on-device on a Raspberry Pi, cutting end-to-end latency below 3 seconds with LIDAR-triggered capture and a MobileNetV2 backbone that eliminated thermal throttling.

CERTIFICATIONS

NPTEL: Applied Accelerated AI; Deep Learning with Computer Vision | **Coursera:** Deep Learning Specialization